

Advances in Neural Retrieval Models for Scholarly Document Processing

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Abstract

This paper presents a comprehensive survey of neural retrieval models applied to scholarly document processing. We examine recent advances in transformer-based architectures for citation extraction, metadata recognition, and full-text indexing of academic publications. Our analysis covers 127 systems published between 2019 and 2024, evaluating their performance on standard benchmarks including S2ORC, DBLP, and CrossRef datasets. We find that hybrid approaches combining traditional pattern matching with learned representations achieve the highest F1 scores on metadata extraction tasks, while end-to-end neural models excel at citation context understanding. We propose a unified evaluation framework and identify key challenges for future research in this rapidly evolving field.

1 Introduction

The exponential growth of scholarly publications poses significant challenges for information retrieval and knowledge management systems. Digital libraries such as Semantic Scholar, Google Scholar, and PubMed index millions of documents annually, requiring robust automated methods for extracting structured metadata from unstructured PDF documents. The transition from rule-based systems to neural approaches has accelerated dramatically in recent years, driven by advances in large language models and multimodal architectures.

Citation extraction and recognition remain fundamental tasks in scholarly document processing. A typical academic paper contains between 20 and 80 references, each encoded in one of hundreds of citation styles. Accurately parsing these references into structured bibliographic records enables downstream applications including citation graph construction, impact measurement, and recommendation systems.

The challenge of PDF processing is compounded by the diversity of document layouts encountered in practice. Conference proceedings, journal articles, technical reports, and preprints each follow different formatting conventions. Two-column layouts, footnotes, headers, and page numbers create complex spatial arrangements that must be correctly interpreted. Furthermore, the distinction between born-digital PDFs with embedded text layers and scanned documents requiring optical character recognition adds another dimension of complexity.

In this survey, we systematically review the state of the art in neural approaches to scholarly document processing. We organize our discussion around three core tasks: (1) metadata extraction from document headers and footers, (2) reference string parsing and matching, and (3) full-text

segmentation and section classification. For each task, we describe the dominant architectural patterns, compare reported results on standard benchmarks, and identify remaining open problems.

Our contributions are as follows. First, we provide a comprehensive taxonomy of neural models for scholarly document processing, covering convolutional, recurrent, transformer-based, and multimodal architectures. Second, we compile and normalize evaluation results across 127 systems, enabling direct comparison of approaches that were previously evaluated on different benchmarks with different metrics. Third, we propose a unified evaluation protocol that addresses current shortcomings in reproducibility and comparability. Fourth, we identify five key challenges that we believe will drive future research in this area.

2 Background and Related Work

2.1 Traditional Approaches

Before the neural era, scholarly document processing relied heavily on rule-based and template-matching approaches. Systems such as ParsCit, GROBID, and CERMINE used conditional random fields (CRFs) and hand-crafted features to segment documents and extract metadata. These systems achieved impressive performance on well-formatted documents but struggled with layout variations and non-standard formatting.

ParsCit, developed by Councill et al. in 2008, introduced a CRF-based approach to reference string parsing that became the de facto standard for over a decade. The system used a rich set of orthographic, lexical, and positional features to label tokens within reference strings as author, title, venue, year, and other fields. Despite its age, ParsCit remains competitive on many benchmarks, particularly for reference strings in standard formats such as APA and IEEE.

GROBID (GeneRation Of Bibliographic Data) took a more comprehensive approach, addressing not only reference parsing but also header extraction, body segmentation, and table recognition. Built on cascaded CRF models, GROBID processes documents in a pipeline architecture, with each stage feeding into the next. The system has been continuously maintained and updated, incorporating neural components in recent versions, and remains one of the most widely deployed open-source solutions for scholarly document processing.

CERMINE (Content ExtRactor and MINEr) combined rule-based layout analysis with machine learning for content classification. Its two-stage architecture first analyzed the geometric structure of document pages using a zone classification approach, then applied CRF models for metadata extraction within identified zones. This separation of layout analysis from content extraction proved effective for handling diverse document formats.

2.2 Neural Approaches

The application of deep learning to document processing began with convolutional neural networks for page segmentation and layout analysis. Early work by Yang et al. (2017) demonstrated that CNNs could effectively classify document regions into semantic categories such as title, abstract, body text, and references, outperforming traditional rule-based approaches on documents with non-standard layouts.

The introduction of recurrent neural networks, particularly bidirectional LSTMs, brought significant improvements to sequence labeling tasks such as reference parsing and named entity recognition in scientific text. Prasad et al. (2018) showed that BiLSTM-CRF models with character-level

embeddings could match or exceed the performance of feature-engineered CRF models while requiring no hand-crafted features.

The transformer revolution, initiated by Vaswani et al. (2017), fundamentally changed the landscape of natural language processing and, by extension, document processing. BERT and its variants provided powerful contextual representations that could be fine-tuned for specific tasks with relatively small amounts of labeled data. SciBERT, a version of BERT pre-trained on scientific text, became the foundation for numerous scholarly document processing systems.

2.3 Multimodal Approaches

Recent work has explored multimodal approaches that combine textual and visual information for document understanding. LayoutLM, developed by Xu et al. (2020), jointly models text, layout, and image features, achieving state-of-the-art results on several document understanding benchmarks. Its successors, LayoutLMv2 and LayoutLMv3, further improved performance by incorporating more sophisticated visual features and pre-training objectives.

DocFormer and SelfDoc represent alternative multimodal architectures that process document images and text simultaneously. These models have shown particular promise for processing documents where layout carries semantic information, such as tables, forms, and multi-column articles.

3 Methodology

3.1 Taxonomy of Approaches

We organize neural approaches to scholarly document processing along four dimensions: architecture type, input modality, training paradigm, and target task. This taxonomy enables systematic comparison of approaches that may appear superficially different but share underlying design principles.

Architecture types include: (a) convolutional networks operating on document images or rendered page regions, (b) recurrent networks processing text sequences with positional features, (c) transformer-based models using self-attention over token sequences, and (d) graph neural networks modeling document structure as graphs with nodes representing text blocks and edges representing spatial or logical relationships.

Input modalities range from text-only (using OCR or embedded text layers) through layout-aware (incorporating bounding box coordinates) to fully multimodal (combining text, layout, and rendered page images). The choice of input modality significantly impacts both performance and computational requirements, with multimodal approaches generally achieving higher accuracy at the cost of increased inference time and memory consumption.

Training paradigms include supervised learning with task-specific labeled data, transfer learning from pre-trained language models, and self-supervised pre-training on unlabeled documents. The availability of large-scale pre-training corpora such as S2ORC (81 million papers) and the IIT-CDIP collection (6 million documents) has made self-supervised approaches increasingly attractive.

3.2 Evaluation Framework

We propose a unified evaluation framework that addresses three key limitations of current evaluation practices. First, we standardize dataset splits and preprocessing procedures to enable direct

comparison across studies. Second, we define a consistent set of metrics for each task, including both token-level and entity-level measures. Third, we establish a protocol for statistical significance testing that accounts for the variance inherent in neural model training.

For metadata extraction, we adopt entity-level precision, recall, and F1 as primary metrics, computed separately for each field (title, author, venue, year, etc.) and macro-averaged across fields. We additionally report exact match accuracy for complete records, as this metric better reflects practical utility.

For reference parsing, we use the same entity-level metrics at the field level, supplemented by complete reference accuracy (the fraction of references where all fields are correctly extracted). We also evaluate linking accuracy, measuring the system’s ability to correctly match parsed references to entries in bibliographic databases.

For full-text segmentation, we report section-level classification accuracy and boundary detection F1. We distinguish between fine-grained segmentation (12 or more section types) and coarse-grained segmentation (5-6 major sections), as the difficulty and practical utility of these tasks differ substantially.

4 Results

4.1 Metadata Extraction

Table 1 summarizes the performance of 42 systems on the metadata extraction task. The highest F1 scores are achieved by hybrid systems that combine neural representations with rule-based post-processing. GROBID-DL, which augments the traditional GROBID pipeline with deep learning components, achieves an overall F1 of 94.7 on the PMC-SA dataset, followed by SciREX at 93.2 and VILA at 92.8.

Pure neural approaches, while generally achieving lower overall F1 scores, show advantages on specific field types. Title extraction, for example, is handled well by transformer-based models due to the relatively consistent formatting of document titles. Author extraction proves more challenging, with name disambiguation and affiliation assignment remaining significant sources of error.

The performance gap between born-digital and scanned documents remains substantial. On born-digital PDFs with clean text layers, the best systems achieve F1 scores above 96 for most metadata fields. On scanned documents processed through OCR, performance drops by 5-15 percentage points depending on scan quality, with author names and reference strings most affected by OCR errors.

4.2 Reference Parsing

Reference parsing results across 53 systems reveal a clear trend toward transformer-based approaches. The current state of the art is held by a fine-tuned SciBERT model with layout features, achieving 97.3 F1 on the CORA dataset and 95.8 on the more challenging UMass dataset.

Interestingly, the choice of citation style has a significant impact on parsing accuracy. Systems trained primarily on author-year formats (APA, Chicago, Harvard) often struggle with numbered citation styles (IEEE, Vancouver), and vice versa. Multi-style training with style-aware attention mechanisms has shown promise in addressing this limitation, with recent work by Kim et al. (2023) demonstrating that style-conditioned models can match or exceed the performance of style-specific models across all tested formats.

4.3 Full-Text Segmentation

Full-text segmentation results show the greatest variation across systems and datasets. The best-performing approach uses a hierarchical transformer architecture that first classifies paragraphs into broad categories (front matter, body, back matter) and then performs fine-grained section classification within each category. This two-stage approach achieves 91.4 accuracy on the PubMed 200K dataset, compared to 88.6 for flat classification.

The identification of section boundaries in documents without explicit section headers remains an open challenge. While most academic papers use numbered sections with descriptive headings, a significant minority employ alternative organizational strategies. Conference papers, in particular, often use minimal formatting, making section boundary detection more difficult.

5 Discussion

5.1 Key Findings

Our analysis reveals several important trends in neural approaches to scholarly document processing. First, the field has converged on transformer-based architectures as the dominant paradigm, with BERT variants serving as the foundation for most recent systems. Second, multimodal approaches incorporating visual features consistently outperform text-only approaches, particularly on documents with complex layouts. Third, hybrid systems that combine neural and rule-based components often achieve the best overall performance, suggesting that domain-specific knowledge remains valuable even in the era of large-scale pre-training.

5.2 Challenges and Future Directions

We identify five key challenges for future research. First, domain adaptation across scientific disciplines remains difficult, as formatting conventions vary significantly between fields. Systems trained on biomedical literature often perform poorly on physics or computer science papers, and vice versa.

Second, the handling of non-English documents receives insufficient attention. Most evaluation benchmarks focus exclusively on English-language publications, yet a significant fraction of the world’s scholarly output is published in other languages. Chinese, Japanese, and Korean documents pose particular challenges due to the absence of whitespace between tokens and the complexity of name formatting conventions.

Third, the processing of supplementary materials, appendices, and online-only content is largely ignored by current systems. As scholarly communication increasingly moves toward enriched digital formats with embedded data, code, and interactive visualizations, document processing systems must evolve to handle these new content types.

Fourth, the computational cost of neural approaches limits their applicability in resource-constrained settings. While cloud-based processing services can handle large-scale document processing, many institutions and researchers require offline processing capabilities. Model compression, distillation, and efficient inference techniques will be critical for making neural document processing widely accessible.

Fifth, the evaluation methodology for document processing systems needs improvement. Current benchmarks often fail to capture the diversity of real-world documents, and evaluation metrics do

not always reflect practical utility. We call for the development of more representative benchmarks and task-specific evaluation protocols.

6 Conclusion

This survey has provided a comprehensive overview of neural approaches to scholarly document processing, covering 127 systems across three core tasks. We have identified key architectural trends, compiled and normalized evaluation results, proposed a unified evaluation framework, and highlighted important challenges for future research.

The field has made remarkable progress in recent years, with neural approaches now achieving near-human performance on well-formatted born-digital documents. However, significant challenges remain in handling scanned documents, non-standard layouts, and multilingual content. We believe that the continued development of multimodal architectures, combined with larger and more diverse pre-training corpora, will drive further improvements in the coming years.

As scholarly communication continues to evolve, document processing systems must adapt to handle new content types and formats. The integration of document processing with knowledge graph construction, scientific question answering, and automated literature review systems represents a particularly promising direction for future work.

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